

Performing Bio-AFM on Live Cells

Live Cell Imaging and Manipulation with Nanosurf Bio-AFM

The study of biological systems ranges from whole organisms, organs and organoids down to their building blocks: cells and proteins. At the lower end of the length scale, atomic force microscope (AFM) systems have found their way into various disciplines of biological research, e.g. imaging of live cells and bacteria, single-cell manipulation, or force spectroscopy on molecules, cells or even tissue. In this application, advanced biological experiments on cells and how certain accessories can be used to perform them are discussed.

Keywords or phrases: Bio-AFM, Atomic Force Microscopy, FluidFM® technology, force spectroscopy, cell stiffness, manipulation.

Introduction

1. Combining Bio-AFM with optical microscopy

Bio-AFM becomes particularly powerful when combined with other techniques, in particular with optical microscopy. For that reason, Nanosurf offers stages for all common brands of inverted microscopes. These stages are mounted to the body of the microscope, allowing the AFM to be placed on top of the bio-sample, between the condenser and objective. There is also a special, long-range condenser for DIC and phase contrast. Epifluorescence, confocal or STED images.

2. Establishing physiological conditions

Mammalian cells are prominent samples for Bio-AFM experiments. These experiments are often conducted at room temperature and with a pH-stabilizing, CO₂-independent buffer. Whereas these non-physiological conditions might be pragmatic and useful to answer

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some scientific problems, they are clearly not optimal. They prevent long-term experiments (> 4 h), as cells visibly deteriorate. Moreover, the validity of some scientific inferences is in question even within short time frames.

It has been shown that buffer conditions affect ion channel conductance¹, molecule uptake², can inflict phototoxicity³ and change cell stiffness⁴, whereas the temperature affects, e.g., cell stiffness⁴ and adhesion measurements⁵. To allow for physiological conditions in an experiment, Nanosurf offers accessories like the AFM-compatible Live Cell Incubator that provide temperature, humidity and CO₂ control.

3. Controlling the temperature

The Petri dish holder accommodates plastic and glass bottom dishes up to 50 mm in diameter and 10 mm in height. It can be used in standalone applications as well as in combination with optical light microscopy. The holder is equipped with heating elements that together with the TEC controller allows controlling the temperature of the sample in a range from ambient to 60 °C. The temperature can be held constant within 0.1 °C. For many cells, ideal proliferation can be ensured when maintaining the temperature at a stable 37 °C.

For temperature-critical experiments, local heating via the Petri dish holder may not be sufficient, because a temperature gradient between the edge of the Petri dish and the center is difficult to prevent. To reduce temperature gradients in the Petri dish, a heating option that elevates the temperature around the complete optical microscope and AFM is available. If the AFM is placed in an acoustic enclosure, heating can be applied to its complete volume.

For long term experiments⁶, a local Petri dish heater and global acoustic enclosure heater were combined, with the global heating targeting 35 °C, just below the target temperature of the Petri dish heater. This combines small gradients with a short response time after mounting a new sample or cantilever.

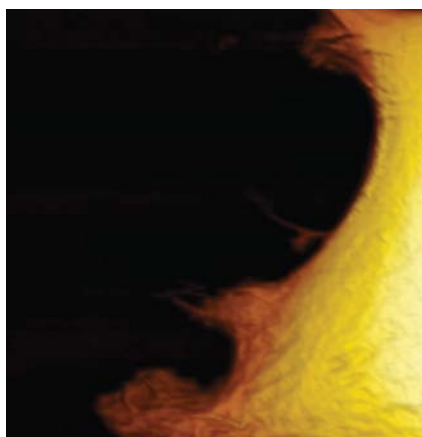
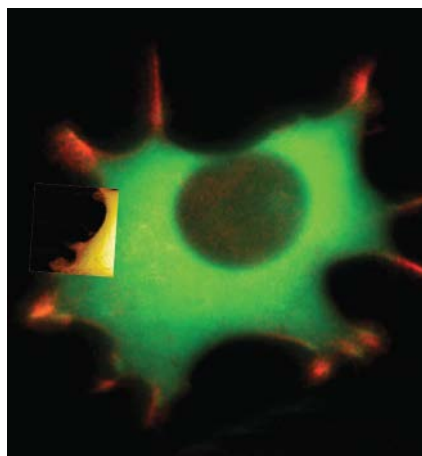


Figure 1: Live fibroblast imaged with inverted light microscope and AFM in cell culture conditions. Top, cell with genetically encoded fluorescent makers (actin-mcherry, gfp-paxillin). Bottom, AFM image, 10 μ m width.

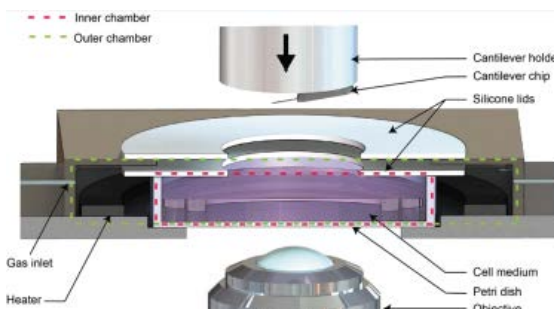


Figure 2: Schematic of the Live Cell Incubator.

Motorization of the AFM, optical microscope and stage mitigate the need to open the acoustic enclosure for laser alignment, finding areas of interest, or changing optical magnification or fluorescence filter set.

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The elevated temperature also increases the evaporation of the medium. To remedy this problem and, additionally, to control the pH of CO₂-dependent cell culture media, Nanosurf offers the Live Cell Incubator option.

4. Controlling humidity and pH

The Live Cell Incubator option is a modification to the standard Petri dish holder. It seals the Petri dish, creating two chambers, while still allowing to freely move the cantilever holder. The inner chamber is comprised of the closed Petri dish while the outer chamber encloses the Petri dish (Figure 2)⁷.

Combining the Live Cell Incubator with a suitable supply of humid gas, evaporation can be reduced to a minimum. The basic approach is to use a washing bottle to humidify bottled gas. In practice, this might be synthetic air with 5% CO₂, which allows to control the pH in CO₂-dependent cell culture media (e.g. DMEM).

Besides the basic humidification scheme, Nanosurf also offers an advanced humidification scheme. The advanced humidification scheme allows to control the humidity from ambient to 95% rel. hum. using a humidity sensor and microcontroller. If so desired, Nanosurf also offers a solution without the requirement for bottled gas, and the mixing of air with CO₂ is done in situ using line gas.

In the fashion described above, the experimental conditions can be kept the same as the cell culture. The correct temperature, low evaporation and pH-control not only ensure relevant insights into near physiological systems but also unlock experiments that unfold over the course of days, as for instance researching tissue formation⁶ with the PicoBalance.

5. Perfusion of cell media

While studying cells under physiological conditions is of great importance, research also requires disturbing the cells in a controlled fashion, e.g. by changing media. As an additional option to the Petri dish holder,

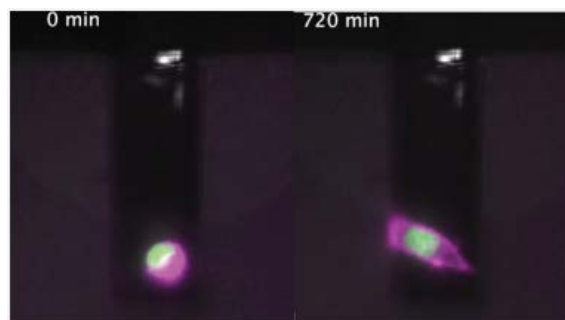


Figure 3: Identical cell at t = 0 and t = 12 h during cell mass measurement (PicoBalance) facilitated by the Live Cell Incubator.

Humidification schemes		
Basic	Advanced	Pro
Bottled gas, ~90% rel. hum.	Bottled gas, ambient to 95% rel. hum.	Contr. gas mixing, ambient to 95% rel. hum.

Table 1: Humidification schemes overview.



Figure 4: Fluid inside the Petri dish can be added or removed using the perfusion insert.

Nanosurf offers the perfusion-insert (Figure 4). With the AFM on top of the Petri dish, ideally also enclosed by the Live Cell Incubator, exchanging media or adding chemicals to the Petri dish becomes more challenging. Particularly, if a certain area of the dish needs to be revisited, or if an experiment is to run autonomously, fluid-exchange or fluid insertion must be possible without human intervention. To allow for this, the perfusion insert consists of a ring, onto which two bent tubes are magnetically attached, which themselves are

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connected to flexible tubing. This tubing can then be connected to, e.g., syringes or pumps, which allows to add a fluid, or exchange media (push-pull-mechanism).

6. Single cell force spectroscopy

The 150 μm -stage is a Petri dish holder, which has long range piezo actuators integrated, allowing to precisely position and move the dish surface in height (z-axis) thus extending the z-range of the AFM. This is of particular interest in the field of single cell force spectroscopy (SCFS) but can also be used for 3D-printing and other applications. In SCFS, cell adhesion can be characterized down to the molecular level. For these experiments, cells can be attached to the cantilever via an adhesive protein and then brought in contact with the dish. After an optional waiting time, they are retracted. Even after the body of the cell is no longer in contact with the surface, adhesion events caused by tethers⁷ are recorded. A positioner for the microscope objective is also available.

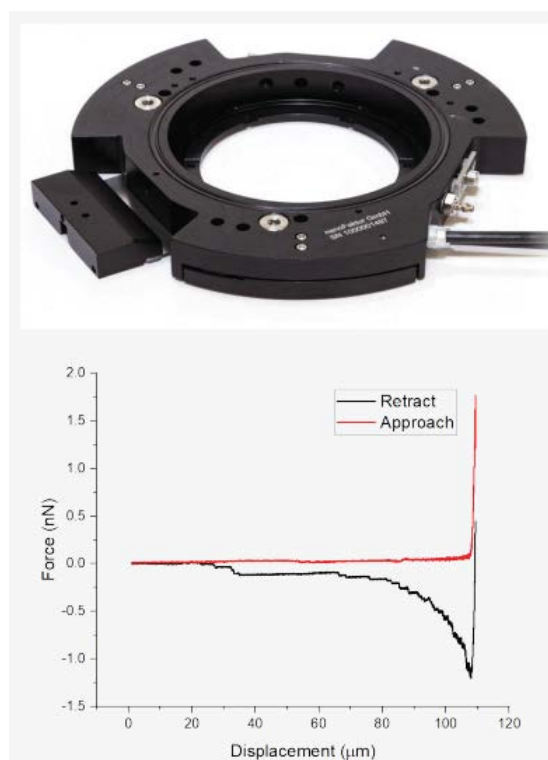


Figure 5: Petri dish holder with integrated piezo motors (top) enabling single cell force spectroscopy experiments (bottom).

7. Cell manipulation via FluidFM[®]

FluidFM combines the functionality of force-sensing cantilevers with that of micro-pipettes, offering a unique possibility of manipulating and probing cells. The FluidFM probe, which is a cantilever with an embedded microfluidic channel and aperture at the free end, allows to partially aspirate the cell, using under pressure. As the holding force between cantilever and cell is considerably higher than using adhesive proteins alone, the adhesion of fully adherent cells can be probed. It also allows for high throughput measurements of single cell adhesion⁹.

Using FluidFM probes that are micro-syringes instead of micropipettes extends FluidFM's functionality even further. These micro-syringes can be used for targeted micro-injection of cells or the reverse, the extraction of cell fluid¹⁰, e.g., for the purpose of TEM, protein assays and PCR or even single cell transcriptomics¹¹.

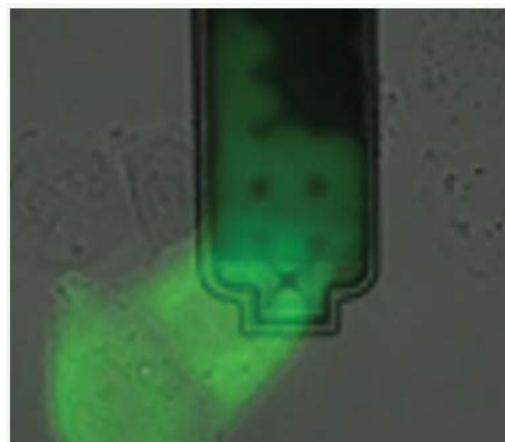


Figure 6: Extraction of sub-picolitre sample of nucleoplasm and cytoplasm from live cells without killing them.

Conclusion

Biological experiments often come with a set of specific requirements. These can relate to establishing a favourable, near-physiological environment in terms of temperature, humidity, and CO₂, or to manipulating the sample, in this case the cell, via the administration of chemicals, separating it from the dish surface. By providing this variety of accessories, in particular the different options for Petri dish holders (heater, Live Cell

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Incubator, perfusion-insert, and piezo-motors), Nanosurf hands researchers the tools to maximize the potential of AFM in biological applications.

List of mentioned accessories

The experiments outlined in this application note were performed with a DriveAFM on a Zeiss inverted optical microscope. Components and accessories used:

- » Petri dish holder with heating
- » Live Cell Incubator
- » Perfusion insert
- » 150- μ m piezo stage
- » FluidFM®

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